EEE4114F: ML Tutorial 2

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Question 1: Neural Networks

Suppose you have a fully-connected, feedforward neural network with two hidden layers. All of the activation functions for each neuron are Rectified Linear Units (ReLU).

The target variables have the form $\boldsymbol{y} = [y_1,\ y_2]$.

The cost function you plan to optimize is given by: $J = \frac{1}{2}(\hat{y_i} - y_i)^2$

Where $\hat{y_i}$ is the predicted output of the model for the i^{th} neuron and y_i is the corresponding target variable.

a. The model is initialised with all bias values set to zero and the weights initialised as follows:

$$W^1 = \begin{bmatrix} 2 \end{bmatrix} \quad W^2 = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

Suppose at a particular moment in time the input is given by $x = 2$, and the target value is given by $y = [1, 2]$.

(i) Calculate the following values based on a single forward pass of the input.

1) $\mathbf{z}^1$, the net weighted inputs into all neurons in the first layer
2) $\mathbf{a}^1$, the activations of neurons in the first layer
3) $\mathbf{z}^2$, the net weighted inputs into all neurons in the second layer
4) $\mathbf{a}^2$, the activations of neurons in the second layer

(ii) After the forward pass has been performed your model parameters need to be updated. Calculate the following

1) $\boldsymbol{\delta}^L$, the errors in the last layer
2) $\boldsymbol{\delta}^1$, the errors in the first hidden layer
3) $\frac{\partial J}{\partial W^1}$, the gradients for weights in the first layer.
4) $\frac{\partial J}{\partial \boldsymbol{b}_1}$, the gradients for the biases in the first layer.

(iii) Using gradient descent calculate the new parameters for W^1 and $\boldsymbol{b}^1$ given that $\eta = 0.1$

Question 2: Reinforcement Learning

You have a mobile robot in the discrete grid world shown in Figure 1. There are five states (locations on the grid) and two possible actions it can take at each time step by moving E/W provided there is space to move in to.

The agent starts in s_3. s_1 and s_5 are terminal states with reward values of $+10$ and $+1$ respectively. The non-terminal states have zero reward values.

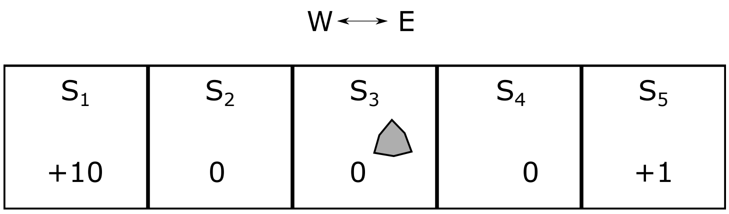

Figure 1: A mobile robot agent in a discrete grid world environment

You decide to use ϵ-greedy Q-learning to train a policy for the agent. All of the initial Q-values are set to zero. The Bellman equation below indicates how the Q-values are updated.

$$Q(s,a) \longleftarrow Q(s,a) + \alpha(N(s,a))[R(s') + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Where $N(s,a)$ is a function that counts the number of times the state-action pair (s,a) has been encountered. The learning rate, $\alpha(n)$ is inversely proportional to $N(s,a)$ such that $\alpha(n) = \frac{1}{n}$. The discount factor is set to $\gamma = 0.9$.

Figure 2 shows the N(s,a) (indicates number of times (s,a) is encountered) and Q(s,a) 'tables' for the second trial/episode. The relative positions of the values in each state cell indicate the action chosen e.g. The number at the left of a cell indicates the action taken is 'W' and to the right of the cell is 'E'. This is to make visualising the transitions and state-action values easier.

W ⟷ E

0 0	0 0	0 2	0 2	2

N(s, a)

0 0	0 0	0 0	0 0.45	0 0

Q(s, a)

Figure 2: The N(s,a) and Q(s,a) values for the second trial/episode t_2

a. How does the variable learning rate provided impact how the agent learns over time?

b. For the third trial the following transitions (state, reward, action) take place based on the action selection policy:

   $(s_3, 0, E) \longrightarrow (s_4, 0, E) \longrightarrow (s_5, 1)$

   Draw the updated N(s,a) and Q(s,a) diagrams after this trial is complete. You may use the same format as Figure 2 or you could write down all of the values in a table if you prefer.

c. Would the agent perform well if it used a greedy action selection policy after t_3? Explain your answer.
d. Would it be a good decision to **decrease** the discount factor, γ, if the robot's decisions do **not** have long-lasting effects in terms of future rewards? Explain your answer.

END